

# Grouped Value Attention: Efficient KV Caching via On-Demand Key Reconstruction

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## Abstract

The KV cache is a primary bottleneck for Transformer decoding: its memory footprint and cache-read traffic grow with sequence length. Grouped-query attention (GQA) reduces this cost by sharing key–value heads, but still stores both a key and a value at every step. We introduce Grouped Value Attention (GVA), which stores grouped values and reconstructs content keys with a learned linear map. At inference, the map can be absorbed into the query, eliminating the need to materialize content keys in the intended decode path. A small shared decoupled RoPE channel retains positional information through a separately cached positional key. For the configurations studied, this representation reduces persistent cache scalars by approximately 45–47% relative to matched GQA. At the 350M-parameter scale with 30B FineWeb-Edu tokens, the 16-dimensional positional variant reaches 44.18 average accuracy across five tasks, compared with 44.36 for GQA and 43.88 for MLA. These results demonstrate near-GQA benchmark accuracy with a more compact cache representation. To translate this compact representation into faster autoregressive inference, we have developed custom decoding kernels and are currently evaluating their end-to-end inference performance with an open-source release planned soon.

## 1 Introduction

The Transformer computes attention from query, key, and value representations [1]. In autoregressive decoding only the newest query is required, while the keys and values of all preceding tokens are reused. Implementations therefore keep a KV cache. Its size grows linearly with context length and often becomes a dominant capacity and bandwidth cost in long-context serving [2, 7].

Grouped-query attention (GQA) reduces this cost by sharing key–value heads across groups of query heads [3]. GQA still writes both a key and a value at every step, so the cache remains two streams. Multi-head latent attention (MLA) compresses keys and values into a joint latent [4]. That shrinks the cache further, at the price of an extra projection and a more involved decode path.

We introduce Grouped Value Attention (GVA). GVA stores only the grouped values and reconstructs keys from them with a learned linear map,  $K = VM$ . Values already carry the content delivered to the attention output; the map selects the features used for scoring. Because the map is fixed at inference, it can be absorbed into the query: the content score against each cached position is a single inner product with the stored value, so the intended decode path need not write a full content-key stream. We find a per-head map more effective than a per-group map.

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Standard RoPE applied to the reconstructed key breaks this absorption. The rotation depends on the cached position and cannot be folded into a single transformed query, requiring either position-dependent key reconstruction or a full key cache. We therefore use a small shared decoupled RoPE channel, following DeepSeek MLA [4]: an unrotated content slice reconstructed from the stored value, plus a short rotated slice shared across heads. Position is retained without restoring a full key cache. Because the positional key is shared, it adds only a few dimensions per token.

The systems case is direct. Token-by-token decoding repeatedly reads the attention cache and is often limited by memory bandwidth. GVA targets faster, more memory-efficient autoregressive inference by reducing the intended persistent cache by approximately 45–47% relative to matched GQA. Prefill remains dominated by parallel matrix operations, so GVA may stay close to GQA there; neither latency nor decode-throughput gains are established by the present experiments. Compared with MLA, GVA uses the value itself as the persistent content state rather than a separate joint latent, preserving a direct value path.

Figure 1 provides an overview of the attention caching strategies and GVA’s value-only content cache.

Our contributions are:

- Grouped Value Attention, which caches grouped values and reconstructs keys with a linear map that is absorbed into the query at inference.
- A small shared decoupled RoPE channel that keeps positional encoding compatible with this reconstruction, at a few extra cached dimensions per token.
- The proposed decoupled-RoPE GVA representation reduces cache scalars by approximately 45–47%; its 16-dimensional positional variant reaches 44.18 average accuracy, close to GQA at 44.36 and MLA at 43.88 with benchmark scores averaged over three runs using different random seeds.

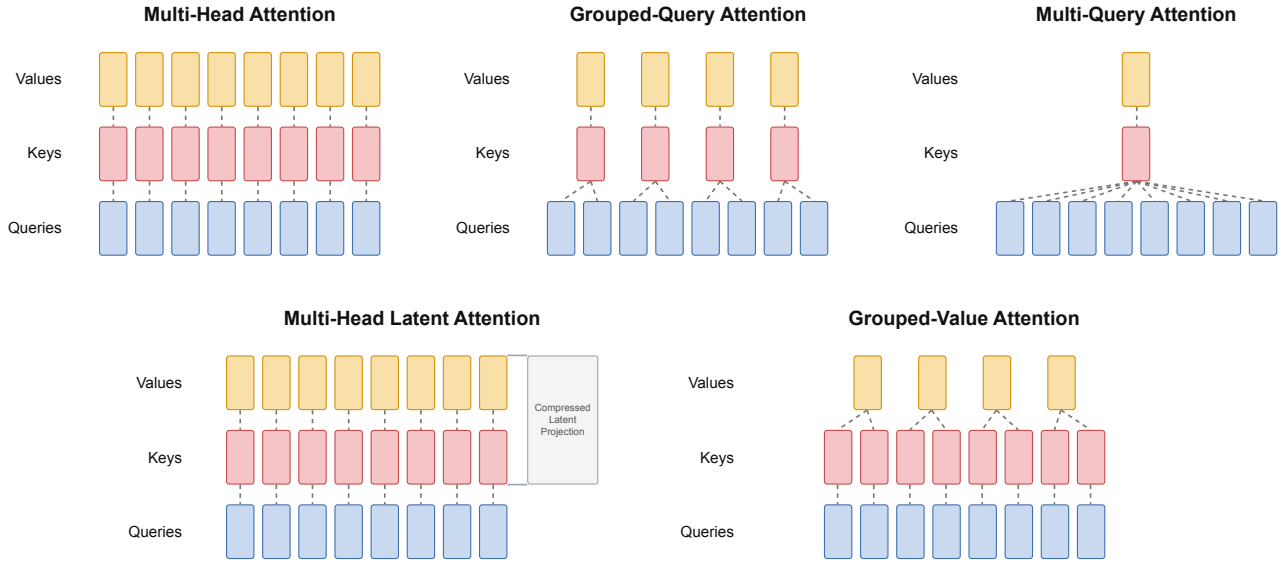


Figure 1: Attention caching strategies for eight query heads. GVA caches four grouped values and reconstructs content keys as  $K = VM$ . The depicted content cache is half the size of matched GQA; including the shared positional key, the total intended cache is approximately 45–47% smaller for the configurations studied.

## 2 Related Work

**Notation.** Let  $H$  be the number of query heads,  $G$  the number of key-value groups,  $d_h$  the head width, and  $T$  the cached length. Per layer and sequence, multi-head attention stores separate keys and values for each head [1, 2]:

$$N_{\text{MHA}} = 2THd_h. \quad (1)$$

**Multi-query attention.** Autoregressive decoding is limited by the bandwidth of loading keys and values at every step [2]. Multi-query attention (MQA) keeps  $H$  query heads but a single shared key-value head ( $G=1$ ) [2]. The cache is

$$N_{\text{MQA}} = 2Td_h, \quad (2)$$

a factor of  $H$  smaller than multi-head attention. That speeds up decoding, at the cost of quality and, in some settings, training stability [2, 3].

**Grouped-query attention.** GQA interpolates between multi-head attention and MQA: query heads are split into  $G$  groups, and each group shares one key-value head [3]. The cache is

$$N_{\text{GQA}} = 2TGd_h. \quad (3)$$

The cases  $G=H$  and  $G=1$  recover multi-head attention and MQA [3]. With a modest  $G$ , GQA achieves quality close to multi-head attention and speed close to MQA in the experiments of Ainslie et al. [3]. It is also used in open models such as Llama 2 70B [16].

**Multi-head latent attention.** DeepSeek MLA compresses keys and values into a low-rank latent of width  $d_c$ , plus a small shared RoPE slice of width  $d_r$ , and caches that instead of full heads [4]:

$$N_{\text{MLA}} = T(d_c + d_r). \quad (4)$$

At decode time the key up-projection can be absorbed into the query projection so that attention need not materialize a full content-key tensor [4]. GVA makes a different choice: the persistent state is the value used in the weighted sum, and only the scoring representation is reconstructed,

$$K = VM. \quad (5)$$

There is no extra latent projection on the cache path. Both designs keep a short RoPE channel outside the absorbable map so position does not force a full key cache. MLA pays for compression with a learned bottleneck; GVA pays with a linear map from value to key.

**Grouped query experts.** Grouped Query Experts (GQE) applies mixture-of-experts routing to the query heads within each GQA group [17]. A router selects query-head experts per token while the key-value heads remain dense and unchanged. GQE therefore reduces active query-head computation while retaining GQA’s cache structure.

## 3 Grouped Value Attention

We hypothesize that the content key can be derived from the value, as both encode information about the same underlying content, eliminating the need for a separate content-key cache.

GVA is built on GQA grouping:  $H$  query heads share  $G$  value heads. Only the key changes. We first reused the value as the key, then replaced that with a linear map from the stored value. We use row vectors throughout this section;  $g(h)$  denotes the value group assigned to query head  $h$ .

### 3.1 Value-only cache and $K = VM$

GQA caches both a grouped key and a grouped value. The two streams have the same shape, so the simplest cut is to store one of them. Values must persist, because they are what attention aggregates.

**Shared KV.** Our first design dropped the key projection and set  $K = V$ . The cache is then exactly half of GQA, but the training loss never recovered to the GQA baseline (Figure 3). One vector is asked both to score and to be retrieved. We trained Lumma-0.6B using this initial shared-KV approach with query normalization, which yielded a slight improvement over shared KV without query normalization, and open-sourced it on Hugging Face as Lumma-0.6B-Base [19].

**Linear reconstruction.** We therefore keep a dedicated value and reconstruct the key,

$$K = VM, \quad M \in \mathbb{R}^{d_h \times d_h}. \quad (6)$$

There is no independent key projection. Only  $V$  is written to the content cache. Values carry the content delivered to the output;  $M$  selects the features used for scoring. We initially compared GVA-H, with one reconstruction map per query head, and GVA-V, with one map per value group that reconstructs a single key shared by the query heads in that group. GVA-H performed better in these initial experiments, so we use it as the default, at a small, sequence-independent parameter cost. We write  $M$  for either choice below, suppressing the head and group indices when discussing a single head.

### 3.2 Absorption at decode

Because  $M$  is fixed at inference, the content key never has to be materialized. The score against a cached position  $j$  is

$$qk_j^\top = q(v_j M)^\top = (qM^\top)v_j^\top = \tilde{q}v_j^\top, \quad (7)$$

where  $\tilde{q} = qM^\top$  is computed once per query token. Content attention then reads only the value cache. This identity is exact. In the fused decode formulation, attention uses  $\tilde{q}$  directly over the value cache, avoiding storage of a temporary key tensor.

Standard RoPE on the reconstructed key breaks the absorption in (7). In row-vector notation,  $k_j = (v_j M)R_j^\top$ , giving  $qk_j^\top = qR_j M^\top v_j^\top$ . The rotation  $R_j$  depends on the cached index  $j$  and sits between  $q$  and  $M^\top$ , so a single transformed query cannot be precomputed for all positions.

### 3.3 Decoupled RoPE

We split each query and key head into an unrotated content slice of width  $d_n$  and a rotated positional slice of width  $d_r$ , while values retain width  $d_h$ . Following DeepSeek MLA [4], RoPE is applied only to the positional slice, and the positional key is shared across heads:

$$k_j^{\text{nope}} = v_j M, \quad M \in \mathbb{R}^{d_h \times d_n}, \quad (8)$$

$$k_j^{\text{rope}} = (x_j W_r)R_j^\top, \quad W_r \in \mathbb{R}^{d_{\text{model}} \times d_r}. \quad (9)$$

The query is split the same way, with a per-head positional slice rotated by  $R_t$  at query position  $t$ . The score is

$$qk_j^\top = q^{\text{nope}}(k_j^{\text{nope}})^\top + q^{\text{rope}}(k_j^{\text{rope}})^\top. \quad (10)$$

The first term still absorbs:  $q^{\text{nope}}(v_j M)^\top = (q^{\text{nope}} M^\top)v_j^\top$ . The second term uses the cached, already-rotated  $k_j^{\text{rope}}$ . Nothing learned sits between the two rotations, so relative position is preserved. Because  $k^{\text{rope}}$  is shared, it costs  $d_r$  scalars per token, not  $Gd_r$ . The full query/key width used for score scaling is  $d_{\text{qk}} = d_n + d_r$ .



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**Algorithm 1** One decode step of GVA with decoupled RoPE

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**Require:** Current hidden state  $x_t$ ; maps  $W_Q, W_V, W_r, \{M_h\}_{h=1}^H$ ; caches  $V_{1:t-1,g}$  for  $g = 1, \dots, G$  and  $k_{1:t-1}^{\text{rope}}$ .

**Ensure:** Per-head attention outputs  $\{o_{t,h}\}_{h=1}^H$ .

```
1:  $q_t \leftarrow x_t W_Q$ ; split each head into  $q_{t,h}^{\text{rope}}, q_{t,h}^{\text{nope}}$ 
2:  $v_t \leftarrow x_t W_V$ 
3: for each value group  $g = 1, \dots, G$  do
4:    $V_{1:t,g} \leftarrow [V_{1:t-1,g}; v_{t,g}]$ 
5: end for
6:  $k_t^{\text{rope}} \leftarrow (x_t W_r) R_t^\top$ 
7:  $k_{1:t}^{\text{rope}} \leftarrow [k_{1:t-1}^{\text{rope}}; k_t^{\text{rope}}]$ 
8: for each query head  $h = 1, \dots, H$  do
9:    $q_{t,h}^{\text{rope}} \leftarrow q_{t,h}^{\text{rope}} R_t^\top$ 
10:   $\tilde{q}_{t,h} \leftarrow q_{t,h}^{\text{nope}} M_h^\top$  {absorb; do not form  $K^{\text{nope}}$ }
11:   $s_{t,h} \leftarrow \tilde{q}_{t,h} V_{1:t,g(h)}^\top + q_{t,h}^{\text{rope}} (k_{1:t}^{\text{rope}})^\top$ 
12:   $o_{t,h} \leftarrow \text{softmax}(s_{t,h} / \sqrt{d_{\text{qk}}}) V_{1:t,g(h)}$ 
13: end for
14: return  $\{o_{t,h}\}_{h=1}^H$ 
```

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### 3.4 Cache size

GQA stores two grouped streams,

$$N_{\text{GQA}} = 2TGd_h. \quad (11)$$

Shared KV and GVA without a positional slice store only values,  $TGd_h$ , exactly half. With decoupled RoPE the persistent state is

$$N_{\text{GVA}} = TGd_h + Td_r, \quad (12)$$

so

$$\frac{N_{\text{GVA}}}{N_{\text{GQA}}} = \frac{1}{2} + \frac{d_r}{2Gd_h}. \quad (13)$$

The second term is a few percent for the widths we use, which is why we say GVA *roughly* halves the GQA cache. Adding positional dimensions on top of the content width changes parameters and the query/key width, not the form of (12):  $V$  stays  $d_h$ -wide and  $k^{\text{rope}}$  stays shared; any change in  $d_r$  is accounted for by the  $Td_r$  term.

For per-group reconstruction, set  $M_h = M_{g(h)}$  in Algorithm 1. The usual concatenation and output projection combine the per-head outputs. Training uses the same content–position split with standard causal attention. Algorithm 1 describes the intended decode path, and the cache count in (12) describes its persistent state rather than measured peak serving memory. We have developed custom decoding kernels and are currently testing their inference performance, with an open-source release planned soon.

## 4 Experiments

We train decoder-only Transformers from scratch at the 350M-parameter scale on a 30B-token sample of FineWeb-Edu [8]. All models use the same data order, token budget, optimizer, and context length unless a variant is named below. The GQA baseline uses grouped key–value heads; MLA and every GVA run keep that query-head count so the comparison is on the cache representation, not on the width of the query.

During pre-training, we used ZClip [18] to mitigate gradient spikes and help prevent loss spikes. ZClip adaptively clips gradients using z-score-based anomaly detection on gradient norms.

## 4.1 Variants

**Shared KV.** The first cache cut sets  $K = V$  and stores only the grouped value. We report it as a reference: the cache is exactly half of GQA, but the loss does not recover (Figure 3). It is not a proposed system.

**GQA and MLA.** GQA caches grouped keys and values [3]. MLA caches a joint latent plus a shared RoPE slice [4]. Both are trained with the same recipe as GVA.

**GVA.** Unless stated otherwise, GVA stores grouped values and reconstructs keys with  $K = VM$ . Our initial experiments compared GVA-H (one map and reconstructed key per query head) with GVA-V (one map and reconstructed key per value group, shared across the query heads in that group). GVA-H performed better, so all variants below use GVA-H unless stated otherwise. We compare:

- **GVA baseline:** linear reconstruction with a standard init of  $M$ .
- **GVA, scale-matched:**  $M$  initialized so  $K$  and  $Q$  start at the same RMS (Appendix A), with query RMSNorm.
- **GVA, variance-fixed:** the same init of  $M$ , without query RMSNorm. This is our default GVA without decoupled RoPE.
- **GVA-H:** one map per query head rather than per KV group.
- **GVA + decoupled RoPE:** the default GVA plus a shared. positional slice of width  $d_r \in \{16, 24\}$  (Section 3.3).

Shared KV and GVA without decoupled RoPE apply ordinary RoPE to the reconstructed key. That path cannot absorb  $M$  at decode; we still report it because it isolates the reconstruction from the positional design.

## 4.2 Evaluation

We report language-model training loss over training steps, and zero-shot accuracy on HellaSwag [9], WinoGrande [10], OpenBookQA [11], and the Easy and Challenge splits of ARC [13]. For each configuration, we perform three runs, each using a different random seed, and report the arithmetic mean of the three accuracies for each task. Average is the unweighted mean of these five task-level means. Small differences should not be interpreted as statistically significant without assessing variability across seeds.

Cache sizes follow Section 3.4: GQA stores  $2TGd_h$ ; shared KV and GVA without a rope slice store  $TGd_h$ ; GVA with decoupled RoPE stores  $TGd_h + Td_r$ .

We have developed custom decoding kernels and are currently evaluating their end-to-end inference performance, including fused decoding throughput, for comparison with MLA and GQA. These systems measurements are not reported here; an open-source release is planned soon.

## 5 Results

**Training loss.** Figure 2 plots language-model loss for GQA, MLA, and the GVA variants. After the initial transient, scale-matched GVA tracks GQA and MLA. The baseline GVA, with standard initialization of  $M$ , is worse early and narrows the gap later, consistent with the scale account in Appendix A. Decoupled-RoPE runs follow the same broad trajectory; we do not observe a second collapse once  $M$  is scale-matched.

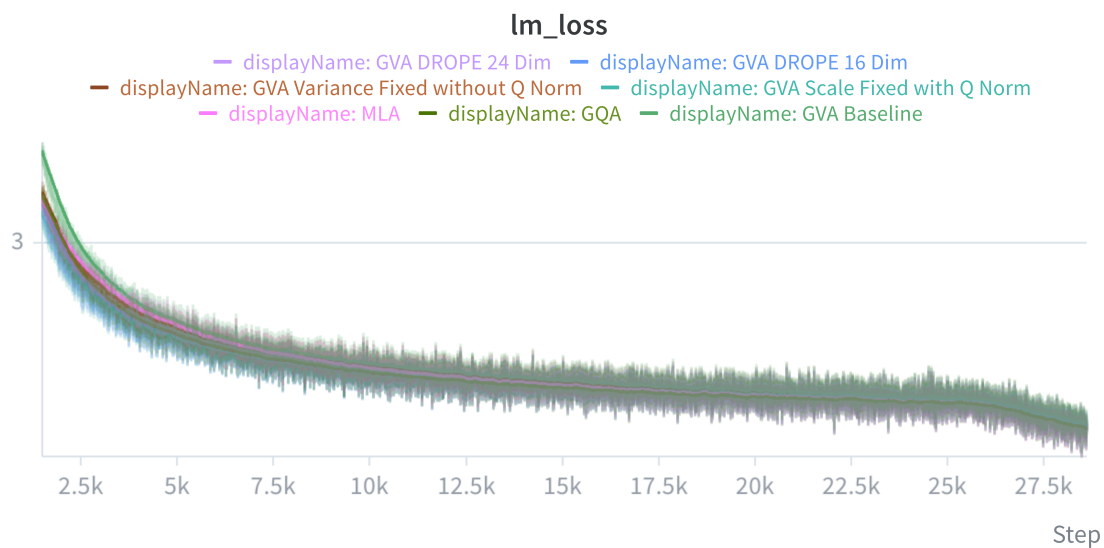


Figure 2: Training loss on FineWeb-Edu for GQA, MLA, and GVA, plotted against training steps. After initialization is scale-matched, GVA follows a similar trajectory to GQA. Shared KV is omitted here; see Figure 3.

Table 1: Accuracy (%) on five benchmarks, averaged over three runs with different random seeds for each configuration. Average is the unweighted mean of the five task-level means. Bold marks the best entry in each column.

| Method                         | HellaSwag    | WinoGrande   | OBQA         | ARC-E        | ARC-C        | Avg.         |
|--------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| GQA                            | <b>43.41</b> | 52.48        | 33.40        | <b>63.42</b> | 29.09        | 44.36        |
| MLA                            | 43.20        | 51.61        | 34.60        | 62.87        | 27.13        | 43.88        |
| GVA baseline                   | 42.12        | 52.96        | <b>34.80</b> | 61.95        | 27.73        | 43.91        |
| GVA, scale-matched + Q-norm    | 42.05        | <b>53.51</b> | 34.40        | 62.75        | <b>29.35</b> | <b>44.41</b> |
| GVA, variance-fixed, no Q-norm | 42.71        | 53.35        | 32.20        | 62.33        | 28.24        | 43.77        |
| GVA + DRoPE $d_r=24$           | 42.94        | 52.40        | 32.10        | 62.29        | 26.79        | 43.30        |
| GVA + DRoPE $d_r=16$           | 42.69        | 53.35        | 33.60        | 62.92        | 28.32        | 44.18        |

**Shared KV.** Figure 3 shows the first cut:  $K = V$  halves the GQA cache, but the loss stays above GQA for the whole displayed run. A single vector does not recover the key/value split. All later GVA runs keep a dedicated value and reconstruct the key.

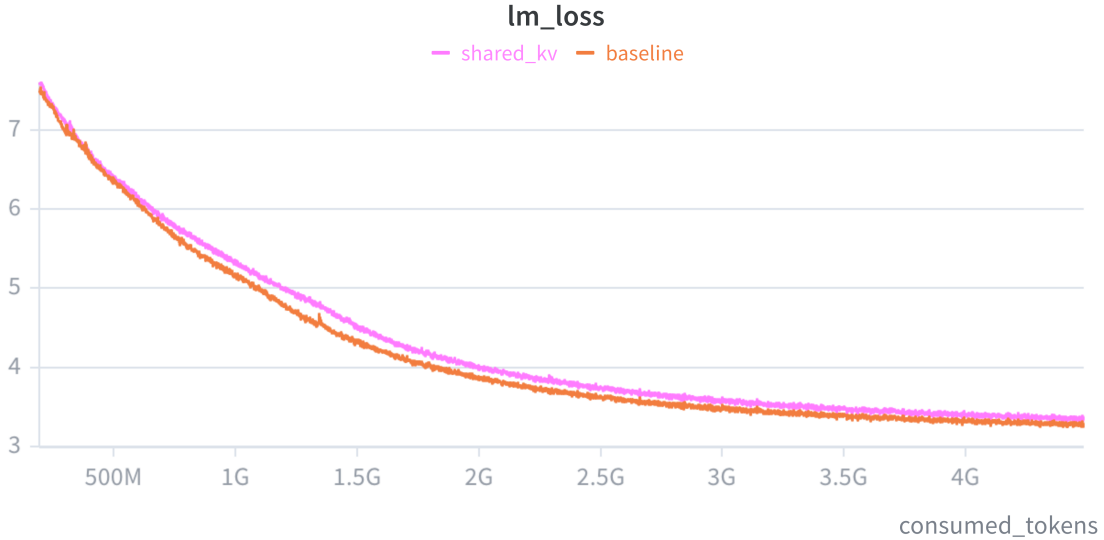


Figure 3: Training loss for shared KV ( $K = V$ ) against the GQA baseline. The cache is half of GQA, but the gap does not close. This motivates reconstructing the key from a dedicated value (Section 3.1).

**Downstream accuracy.** Table 1 reports zero-shot accuracy averaged over three runs with different random seeds for each configuration. Average is the unweighted mean of the five task-level mean accuracies.

Scale-matched GVA with query RMSNorm is the strongest GVA row (44.41) and sits close to GQA (44.36). The default without query RMSNorm is slightly lower (43.77) but remains close to MLA (43.88). The GVA baseline without matched initialization reaches 43.91: reconstruction alone is already in this band, and scale matching primarily improves early loss rather than consistently improving the final average across variants.

Decoupled RoPE is the proposed serving design. With  $d_r=16$ , the average is 44.18, 0.18 percentage

points below GQA. With  $d_r=24$ , it is 43.30. We treat  $d_r=16$  as the better operating point among the two widths tested. These runs do not isolate the effects of positional width and content-width allocation. Neither DRoPE row beats GQA on the average; the claim is a much smaller intended cache at near-GQA quality, not a quality win.

**Cache.** Relative to GQA, shared KV and GVA without a separate positional slice require half the cache scalars when only values are retained. With DRoPE the ratio is  $\frac{1}{2} + d_r/(2Gd_h)$  (Section 3.4), corresponding to about 47% saved at  $d_r=16$  and 45% at  $d_r=24$  for our grouping. These are representation-level counts, not measured serving-memory reductions. Ordinary-RoPE variants cannot use the absorbed decode path and require position-dependent key reconstruction.

## 6 Limitations

The roughly 45–47% cache reduction for decoupled-RoPE GVA describes the *intended* decode state: grouped  $V$  plus a shared  $k^{\text{rope}}$ . Omitting the positional slice gives the idealized 50% value-only reduction. We have developed custom decoding kernels and are currently testing their inference performance, with an open-source release planned soon. Fused decode throughput, peak serving memory, and batch capacity measurements are not reported here.

All reported comparison runs use one scale (approximately 350M parameters), one data mix (30B FineWeb-Edu tokens), and three different random seeds per benchmark configuration, with scores averaged across runs. DRoPE with  $d_r=24$  is behind GQA on average; we have not systematically swept RoPE width, additive versus carved allocation, or longer contexts. Shared KV is a failed first cut, not a baseline we recommend.

## 7 Conclusion

GVA stores grouped values and reconstructs keys with a linear map  $K = VM$ . Sharing one vector as both key and value halves the cache but does not match GQA in the observed run. Reconstructing the key and matching its initial scale to the query yields quality close to GQA. A small shared decoupled RoPE channel keeps position compatible with absorbing  $M$  at decode. Relative to GQA, the intended persistent cache is roughly half, while downstream quality remains in the same broad band (Table 1). We have developed custom decoding kernels and are currently testing their end-to-end inference performance, with an open-source release planned soon. Further work includes completing this evaluation and a broader sweep of RoPE width, model scale, and random seeds.

## References

- [1] A. Vaswani et al. Attention Is All You Need. *NeurIPS*, 2017.
- [2] N. Shazeer. Fast Transformer Decoding: One Write-Head is All You Need. *arXiv:1911.02150*, 2019.
- [3] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebrón, and S. Sanghai. GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints. *EMNLP*, 2023.
- [4] DeepSeek-AI. DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model. *arXiv:2405.04434*, 2024.
- [5] B. Zhang and R. Sennrich. Root Mean Square Layer Normalization. *NeurIPS*, 2019.

- [6] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. *NeurIPS*, 2022.
- [7] W. Kwon et al. Efficient Memory Management for Large Language Model Serving with PagedAttention. *SOSP*, 2023.
- [8] G. Penedo, H. Kydlíček, L. Ben Allal, A. Lozhkov, M. Mitchell, C. Raffel, L. von Werra, and T. Wolf. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. *arXiv:2406.17557*, 2024.
- [9] R. Zellers, A. Holtzman, Y. Bisk, A. Farhadi, and Y. Choi. HellaSwag: Can a Machine Really Finish Your Sentence? *ACL*, 2019.
- [10] K. Sakaguchi, R. Le Bras, C. Bhagavatula, and Y. Choi. WinoGrande: An Adversarial Winograd Schema Challenge at Scale. *AAAI*, 2020.
- [11] T. Mihaylov, P. Clark, T. Khot, and A. Sabharwal. Can a Suit of Armor Conduct Electricity? A New Dataset for Open Book Question Answering. *EMNLP*, 2018.
- [12] Y. Bisk, R. Zellers, R. Le Bras, J. Gao, and Y. Choi. PIQA: Reasoning about Physical Commonsense in Natural Language. *AAAI*, 2020.
- [13] P. Clark et al. Think You Have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge. *arXiv:1803.05457*, 2018.
- [14] Z. Liu et al. KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache. *ICML*, 2024.
- [15] G. Hinton, O. Vinyals, and J. Dean. Distilling the Knowledge in a Neural Network. *arXiv:1503.02531*, 2015.
- [16] H. Touvron et al. Llama 2: Open Foundation and Fine-Tuned Chat Models. *arXiv:2307.09288*, 2023.
- [17] V. Tripathi and A. Kumar. Grouped Query Experts: Mixture-of-Experts on GQA Self-Attention. *arXiv:2606.20945*, 2026.
- [18] A. Kumar, L. Owen, N. Roy Chowdhury, and F. Göra. ZClip: Adaptive Spike Mitigation for LLM Pre-Training. *arXiv:2504.02507*, 2025.
- [19] FrontiersMind. Lumma-0.6B-Base. *Hugging Face model repository*, 2026. <https://huggingface.co/FrontiersMind/Lumma-0.6B-Base>.

## A Query–Key Scale

If query and key coordinates are centered and weakly correlated, with standard deviations  $\sigma_Q$  and  $\sigma_K$ , then for row vectors

$$\text{Var}\left(\frac{qk^\top}{\sqrt{d_h}}\right) \approx \sigma_Q^2 \sigma_K^2. \quad (14)$$

When the product is too small, softmax sees nearly equal logits and attention is flat. When it is too large, a few positions dominate. Figure 4 shows both regimes.

GQA obtains  $Q$  and  $K$  from separate projections of the same hidden state, providing a reference for their initial scales rather than a guarantee that they match. In GVA,  $K = VM$ , so the scale of  $K$  also depends on  $M$ . In our initial runs, default initialization left  $K$  far below  $Q$ , and early training was spent adjusting this mismatch.

**Initialization.** Let  $d_{\text{in}}$  be the input width of  $M$ , namely the value width. Assuming approximately centered value coordinates with variance  $\sigma_V^2$ , independent of the initial map entries, we draw

$$M_{ij} \sim \mathcal{N}\left(0, \frac{\sigma_Q^2}{\sigma_V^2 d_{\text{in}}}\right). \quad (15)$$

Then  $\text{Var}(K_j) \approx d_{\text{in}} \sigma_V^2 \text{Var}(M_{ij}) = \sigma_Q^2$ , so  $K$  starts at approximately the same RMS as  $Q$ . The target uses the query scale after any query normalization. After initialization,  $M$  remains learnable. Optional query RMSNorm [5] controls the query scale, but the variance-fixed row in Table 1 remains competitive without it. This initialization does not normalize the cached values.

**Decoupled RoPE.** Apply the same rule per slice. The fan-in of  $M$  is the value width, not the content width. Initialize the shared positional-key projection so  $k^{\text{rope}}$  matches the initial RMS of  $q^{\text{rope}}$ . Assess positional and content scales separately: pooling them into one key variance hid the mismatch in early runs.

Figure 5 complements Figure 4 by varying query and key scales together, showing that equal scales alone do not prevent flat or highly concentrated attention.

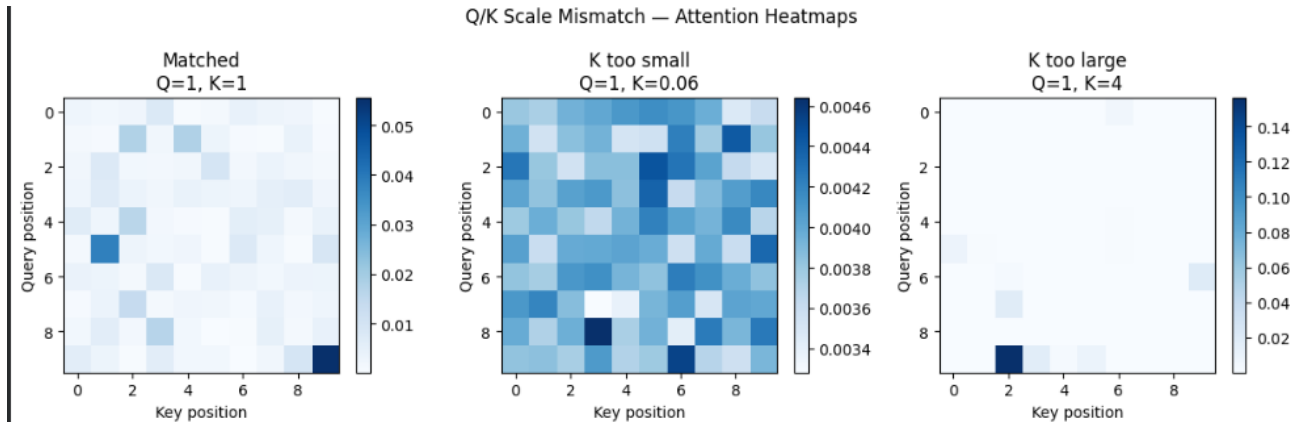


Figure 4: Illustrative attention heatmaps on a 10-position sequence with matched query and key scales (left), keys that are too small (middle), and keys that are too large (right). Small keys produce nearly uniform attention; large keys concentrate the weights. Each panel uses its own color scale.



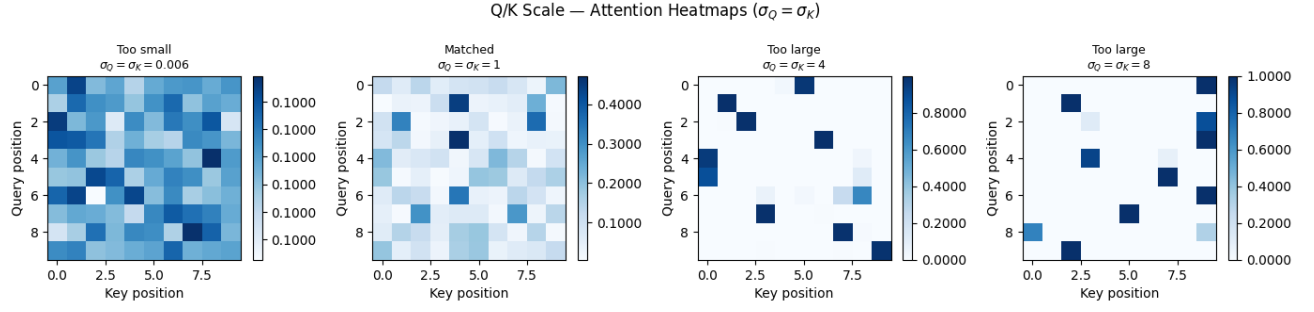


Figure 5: Attention heatmaps ( $10 \times 10$ ) with matched scales  $\sigma_Q = \sigma_K \in \{0.006, 1, 4, 8\}$  (left to right). Small  $\sigma$  gives nearly uniform weights ( $\approx 0.1$  per key); unit  $\sigma$  gives moderate variation; large  $\sigma$  gives peaked, near one-hot rows. Each panel uses its own color scale. Absolute scale controls whether softmax is flat, selective, or peaked.

## B Extra Evaluation Curves

Per-task accuracy over training for the seven configurations in Table 1. The plots retain the original run names and horizontal-axis units.

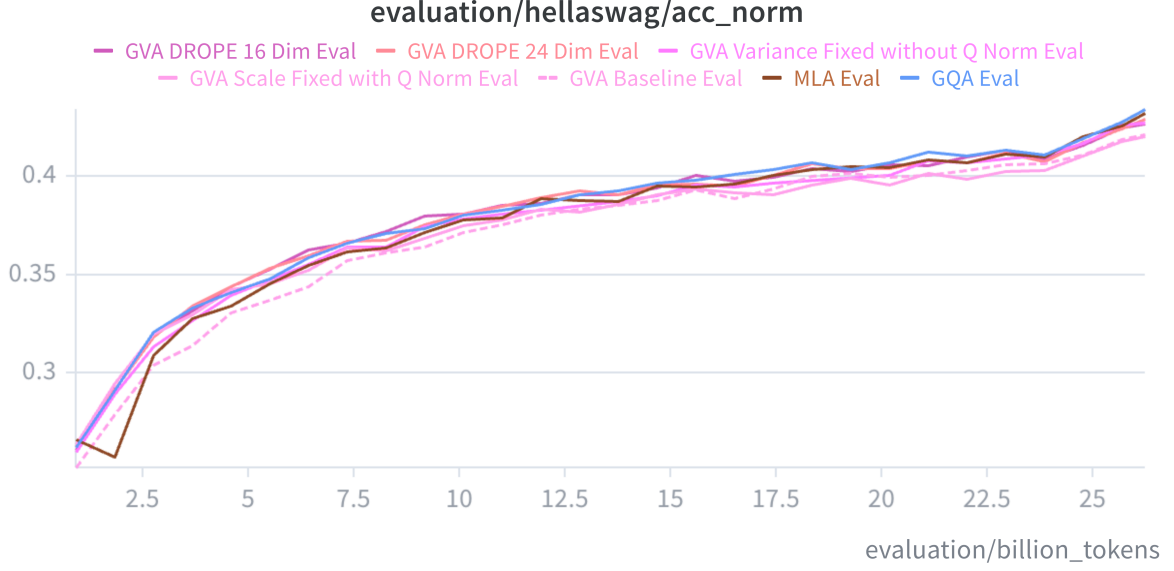


Figure 6: HellaSwag accuracy over training.

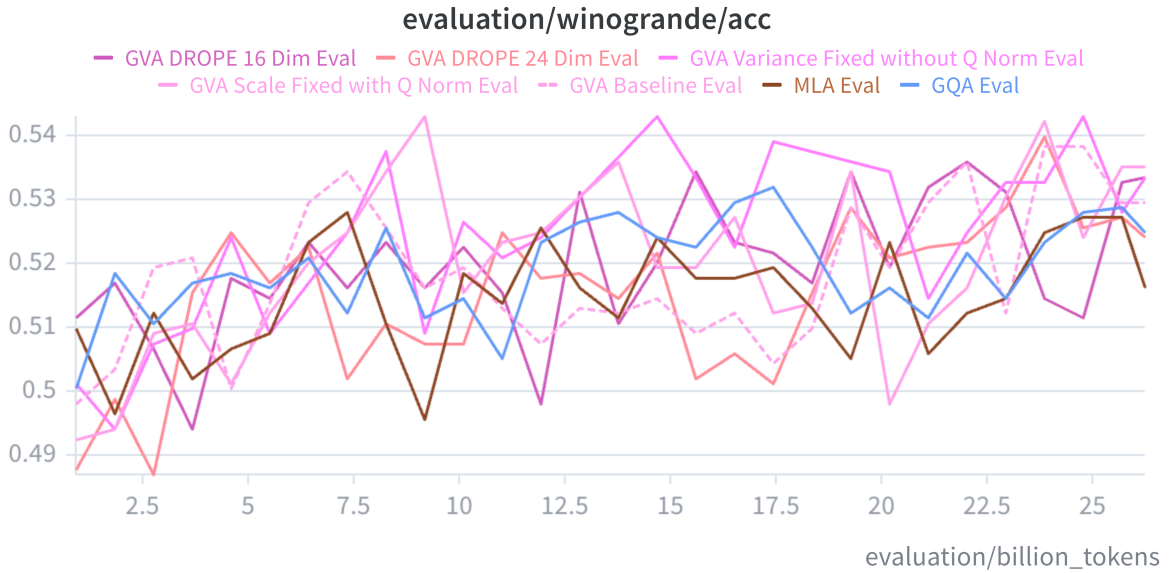


Figure 7: WinoGrande accuracy over training.

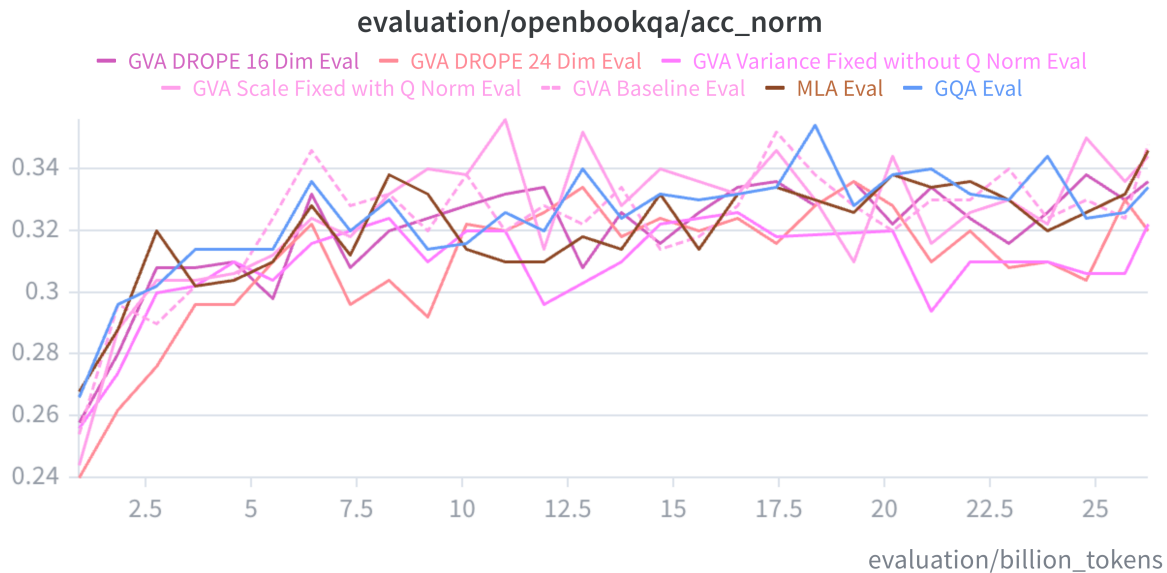


Figure 8: OpenBookQA accuracy over training.

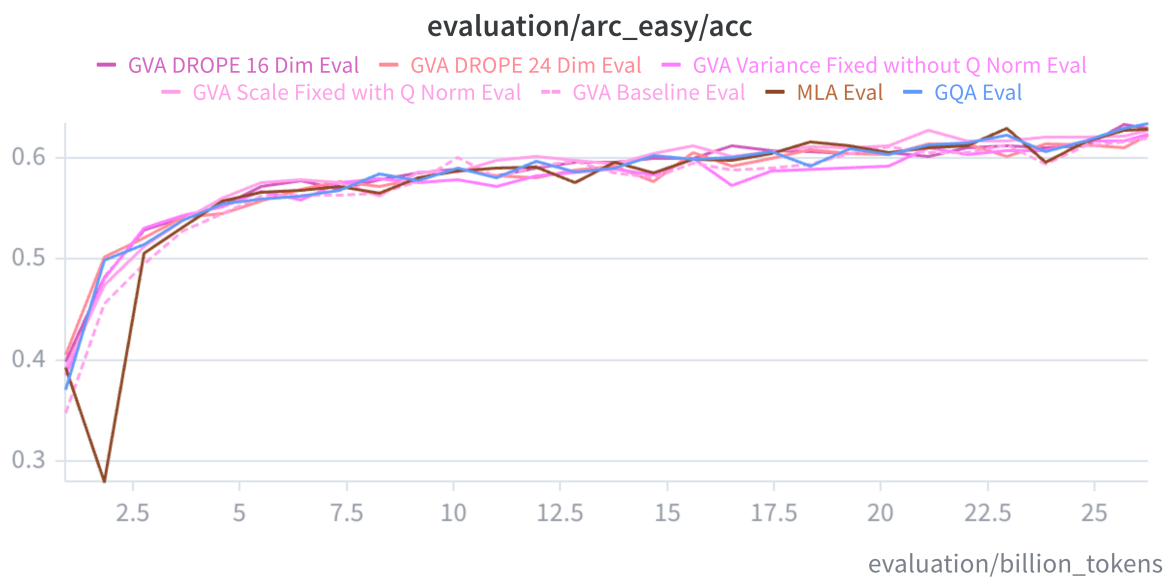


Figure 9: ARC-Easy accuracy over training.

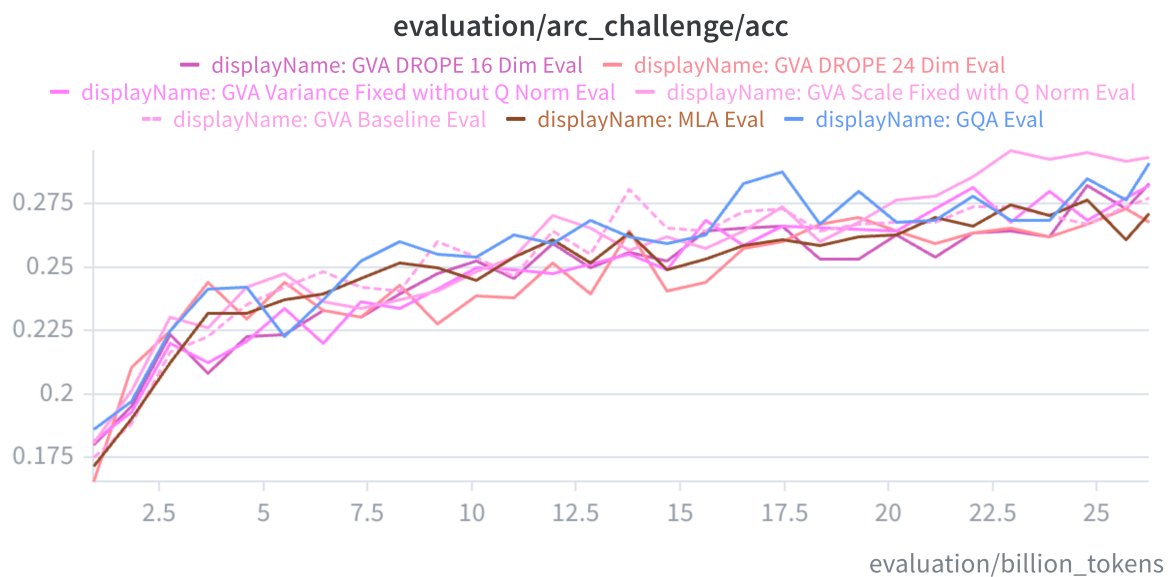


Figure 10: ARC-Challenge accuracy over training.